

SQL Task

1) Local Variable (DECLARE)

Task: Declare a local variable @MinSalary = 1500 and return all employees whose Salary >= @MinSalary,

showing: Fname, Lname, Salary.

2) Local Variable + Aggregation

Task: Declare @DeptNo = 30 then return:

- number of employees in that dept
 - average salary in that dept
- Output columns: EmpCount, AvgSalary.

3) Global variable (system global variable like @@ROWCOUNT)

Task: Select all employees in department 20, then immediately return how many rows were returned using @@ROWCOUNT.

4) CASE WHEN (Conditional Statement using CASE END)

Task: Return each employee with a new column SalaryLevel:

- High if Salary >= 2000
- Mid if Salary between 1000 and 1999
- Low if Salary < 1000

Output: Fname, Lname, Salary, SalaryLevel.

5) CASE END (Handle NULL department)

Task: Return employees with Dno and a column DeptStatus:

- No Department if Dno is NULL
- Has Department otherwise

Output: Fname, Lname, Dno, DeptStatus.

6) IIF condition (simple)

Task: Using IIF, return a column HasSupervisor:

- Yes if Superssn is NOT NULL
- No otherwise

Output: Fname, Lname, Superssn, HasSupervisor.

7) IF condition (control-of-flow)

Task: Declare @DeptNo = 10.

If there are employees in that department → return them.

Else → return a single message: "No employees in this department".

8) WHILE loop (insert/update using loop)

Task: Declare @Inc = 50 and @TargetSalary = 2000.

While there exists any employee with Salary < @TargetSalary, increase those employees' salary by @Inc.

After the loop, return the final list: Fname, Lname, Salary.

Goal: practice WHILE, UPDATE, and loop stopping condition.